

EKAM MANN

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Open to Relocation

EDUCATION

The University of British Columbia

Bachelor of Applied Science in Mechanical Engineering

- 3.8 GPA | Dean's List | RCAF Foundation Scholar

Vancouver, BC

Expected May 2029

EXPERIENCE

Petra Engineers and Consultants Ltd.

Mechanical Design Engineering Intern

Calgary, AB

May 2026 – August 2026

- Calculated minimum required wall thickness for 12 pipe segments using a CSA Z662 structural sizing spreadsheet, producing sizing summaries used directly in construction drawings.
- Performed pipe stress analysis in CAEPIPE for a natural gas distribution system, interpreting code stress results against allowables to identify support locations requiring senior engineering review.
- Redlined 4 engineering drawing packages against ASME B31.3, CSA Z662, and ASME B16.5 requirements, with 11 review comments incorporated into issued revisions.

UBC Solar

Aeroshell Designer

Vancouver, BC

September 2025 – Present

- Reduced aeroshell drag coefficient 37.5% (C_d 0.176 to 0.110) while managing a lift coefficient tradeoff (C_l -1.0 to -0.55), through iterative geometry redesign validated with steady RANS ($k-\omega$ SST) CFD simulations.
- Authored fully toleranced GD&T drawings in line with ASME Y14.5 for 4 aeroshell parts released for fabrication, by defining datum schemes and feature control frames to control fitment between mating components in SolidWorks.
- Contributed to 3 of 7 aeroshell design iterations toward a manufacturing-ready release by modelling topshell and canopy geometry in SolidWorks surfacing, incorporating iterative design feedback and maintaining revision-controlled CAD outputs.

ENGINEERING PROJECTS

Polycarbonate Windshield: Design, Thermoforming, and Failure Analysis | UBC Solar

- Chose vacuum thermoforming over machined polycarbonate to cut estimated part cost from \$2,000 to \$300, by designing and building a waterjet-cut forming jig for use in an industrial oven.
- Achieved approximately 2.4% drag reduction over baseline per CFD, by modelling a double-curvature polycarbonate windshield in SolidWorks to improve laminar flow transition between canopy and aeroshell.
- Controlled panel-to-panel fitment across the canopy-to-aeroshell interface, by designing the CFRP L-flange attachment geometry and eliminating a bondline stress concentration under peak thermal loading through FEA-based failure analysis.
- Evaluated Sikaflex 252 against MS polymer, VHB tape, and neutral-cure silicone alternatives to select the polycarbonate-to-CFRP bonded joint system for thermal expansion compatibility and long-term serviceability.

Vortex Generator Research: NACA 2412 Stall Characteristics | Grant-Funded Independent Research

- Improved stall angle of a NACA 2412 airfoil by 76%, by designing and wind tunnel testing 14 vortex generator configurations.
- Enabled all experimental runs for under \$50, by designing and building a wind tunnel from scratch.

Autonomous Underwater Claw | Course Project

- Automated grip, lift-confirmation, and retry without operator input, by implementing a four-state control machine in C/C++ on Arduino driving a 180° servo from ultrasonic range feedback.
- Diagnosed intermittent false triggering to overlapping open-loop threshold conditions causing state chatter, then eliminated it by adding closed-loop lift verification and a rolling-average filter.
- Reduced material waste 20% pre-build, by simulating interference envelopes in SolidWorks and fabricating sheet-metal components to ± 2 mm tolerances across 12 joints.

TECHNICAL SKILLS

CAD and Simulation: SolidWorks (Surfacing, FEA), AutoCAD, ANSYS CFD, CAEPIPE

Design and Analysis: GD&T (ASME Y14.5), Tolerance Analysis, Structural Sizing, DFM/DFA, FMEA

Manufacturing: CNC Machining, Manual Lathe and Mill, Waterjet Cutting, Vacuum Thermoforming, Composite Layup, FDM 3D Printing

Programming and Controls: MATLAB, Python, Arduino (C/C++), Closed-Loop Control, MS Excel

Codes and Standards: ASME Y14.5, ASME B31.3, ASME B16.5, CSA Z662